

L28 Section 8.4

Theorem. Existence of Analytic Solutions

Suppose x_0 is an ordinary point for equation

$$y'' + P(x)y' + Q(x)y = 0. \quad \rightarrow P, Q \text{ analytic at } x_0$$

Then the equation has two linearly independent analytic solutions of the form

$$y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n.$$

Moreover, a power series solution converges at least on some interval $|x - x_0| < R$, where R is the distance from x_0 to the closest singular point of the equation in the complex plane.

P, Q $\uparrow$ not analytic.

ex. Find the minimum radius of convergence of a power series solution of the equation

$$(x^2 - 2x + 5)y'' + xy' - y = 0 \Rightarrow y'' + \frac{x}{x^2 - 2x + 5}y' - \frac{1}{x^2 - 2x + 5}y = 0$$

1. about the ordinary point $x = 0$.

where are the singular points? $x^2 - 2x + 5 = 0$

$$\Rightarrow x = \frac{2 \pm \sqrt{4 - 4(5)(1)}}{2} = \frac{2 \pm \sqrt{-16}}{2}$$

distance $(0,0) \rightarrow (1,2)$

$$r = \sqrt{(0-1)^2 + (0-2)^2} = \sqrt{5}$$

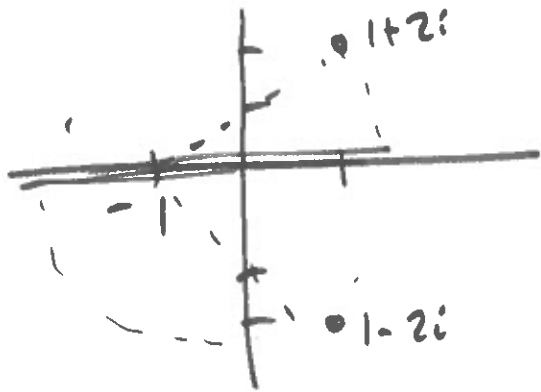
$$\frac{2 \pm 4i}{2} = 1 \pm 2i$$

singular points

distance between $a+ib, c+id$ $(a,b), (c,d)$

$$\sqrt{(a-c)^2 + (b-d)^2}$$

2. about the ordinary point $x = -1$. $(-1, 0)$ $(1, 2)$



$$\begin{aligned} \text{distance } r &= \sqrt{(-1-1)^2 + (0-2)^2} \\ &= \sqrt{4+4} = \sqrt{8} \end{aligned}$$

guaranteed convergence on
 $(-1-\sqrt{8}, -1+\sqrt{8})$

ex. Find the first three nonzero terms in a power series expansion about $x = 1$ for a general solution to

$$2y'' + xy' + y = 0.$$

no singular points

solution's radius of convergence: ∞ .

$$y = \sum_{n=0}^{\infty} a_n (x-1)^n, \quad y' = \sum_{n=1}^{\infty} n a_n (x-1)^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n (x-1)^{n-2}$$

$$2y'' + xy' + y = 2 \sum_{n=2}^{\infty} n(n-1) a_n (x-1)^{n-2} + x \sum_{n=1}^{\infty} n a_n (x-1)^{n-1} + \sum_{n=0}^{\infty} a_n (x-1)^n = 0$$

not centered at 1.

$$x = (x-1) + 1$$

$$\Rightarrow \sum_{n=2}^{\infty} 2n(n-1) a_n (x-1)^{n-2} + \sum_{n=1}^{\infty} n a_n (x-1)^n + \sum_{n=1}^{\infty} n a_n (x-1)^{n-1} + \sum_{n=0}^{\infty} a_n (x-1)^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} 2(n+2)(n+1) a_{n+2} (x-1)^n + \sum_{n=1}^{\infty} n a_n (x-1)^n + \sum_{n=0}^{\infty} (n+1) a_{n+1} (x-1)^n + \sum_{n=0}^{\infty} a_n (x-1)^n = 0$$

$$\Rightarrow (2(2 \cdot 1) a_2 + 1 \cdot a_1 + a_0) + \sum_{n=1}^{\infty} [2(n+2)(n+1) a_{n+2} + (n+1) a_{n+1} + (n+1) a_n] (x-1)^n = 0$$

$$\Rightarrow \begin{cases} 4a_2 + a_1 + a_0 = 0 & (n=0) \\ 2(n+2)(n+1) a_{n+2} + (n+1) a_{n+1} + (n+1) a_n = 0, & n \geq 1 \end{cases}$$

$$\Rightarrow a_{n+2} = \frac{-\cancel{(n+1)}a_{n+1} - \cancel{(n+1)}a_n}{2(n+2)\cancel{(n+1)}} = \frac{-a_{n+1} - a_n}{2(n+2)}$$

a_0, a_1 constants

$$n=0: 4a_2 + a_1 + a_0 = 0 \Rightarrow 4a_2 = -a_1 - a_0 \Rightarrow a_2 = -\frac{1}{4}a_1 - \frac{1}{4}a_0$$

$$n=1: a_3 = \frac{-a_2 - a_1}{2(3)} = \frac{-(-\frac{1}{4}a_1 - \frac{1}{4}a_0) - a_1}{6} = \dots = -\frac{1}{8}a_1 + \frac{1}{24}a_0$$

$$y = \sum_{n=0}^{\infty} a_n (x-1)^n = a_0 + a_1(x-1) + a_2(x-1)^2 + a_3(x-1)^3 + \dots$$

$$= a_0 + a_1(x-1) + \left(-\frac{1}{4}a_1 - \frac{1}{4}a_0\right)(x-1)^2 + \left(-\frac{1}{8}a_1 + \frac{1}{24}a_0\right)(x-1)^3 + \dots$$

$$= a_0 \left(1 - \frac{1}{4}(x-1)^2 + \frac{1}{24}(x-1)^3 + \dots \right)$$

$$+ a_1 \left((x-1) - \frac{1}{4}(x-1)^2 - \frac{1}{8}(x-1)^3 + \dots \right)$$

ex. Find at least the first four nonzero terms in a power series expansion for $y'' + (3x - 1)y' - y = 0$ $x_0 = -1$.

$$y = \sum_{n=0}^{\infty} a_n (x+1)^n$$

not centered
at -1

$$3x - 1 = a(x+1) + b$$

$$\Rightarrow 3(x+1) - 4$$

$$y'' + (3(x+1) - 4)y' - y = \sum_{n=2}^{\infty} n(n-1)a_n(x+1)^{n-2} + (3(x+1) - 4) \sum_{n=1}^{\infty} n a_n(x+1)^{n-1} - \sum_{n=0}^{\infty} a_n(x+1)^n = 0$$

$$\Rightarrow \sum_{n=2}^{\infty} \underbrace{n(n-1)a_n(x+1)^{n-2}}_{\uparrow 2} + \sum_{n=1}^{\infty} 3n a_n(x+1)^{n-1} - \sum_{n=1}^{\infty} \underbrace{4n a_n(x+1)^{n-1}}_{\uparrow 1} - \sum_{n=0}^{\infty} a_n(x+1)^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}(x+1)^n + \sum_{n=1}^{\infty} 3n a_n(x+1)^{n-1} - \sum_{n=0}^{\infty} 4(n+1)a_{n+1}(x+1)^n - \sum_{n=0}^{\infty} a_n(x+1)^n = 0$$

$$\Rightarrow (2a_2 - 4a_1 - a_0) + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} - 4(n+1)a_{n+1} + (3n-1)a_n](x+1)^n = 0$$

$$\Rightarrow \begin{cases} 2a_2 - 4a_1 - a_0 = 0 & (n=0) \\ a_{n+2} = \frac{4(n+1)a_{n+1} - (3n-1)a_n}{(n+2)(n+1)}, & n \geq 1 \end{cases}$$

$$a_0, a_1$$

$$2a_2 - 4a_1 - a_0 = 0 \Rightarrow a_2 = 2a_1 + \frac{1}{2}a_0$$

$$n=1: a_3 = \frac{8a_2 - 2a_1}{6} = \frac{8(2a_1 + \frac{1}{2}a_0) - 2a_1}{6} = \frac{14a_1 + 4a_0}{6}$$

$$\Rightarrow a_3 = \frac{7}{3}a_1 + \frac{2}{3}a_0$$

$$\text{5. } y = \sum_{n=0}^{\infty} a_n (x+1)^n = a_0 + a_1(x+1) + (2a_1 + \frac{1}{2}a_0)(x+1)^2 + (\frac{7}{3}a_1 + \frac{2}{3}a_0)(x+1)^3 + \dots$$

$$= a_0 \left(1 + \frac{1}{2}(x+1)^2 + \frac{2}{3}(x+1)^3 + \dots \right)$$

$$+ a_1 \left((x+1) + 2(x+1)^2 + \frac{7}{3}(x+1)^3 + \dots \right)$$

ex. Find at least the first four nonzero terms in a power series expansion for $y' - e^x y = 0$ $y(0) = 1$ $\rightarrow a_0 = 1$

center

$$y' - e^x y = \sum_{n=1}^{\infty} n a_n x^{n-1} - e^x \sum_{n=0}^{\infty} a_n x^n = 0$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$= \sum_{n=1}^{\infty} n a_n x^{n-1} - \left(\sum_{n=0}^{\infty} \frac{x^n}{n!} \right) \left(\sum_{n=0}^{\infty} a_n x^n \right) = 0$$

↑
centered at 0.

$$= (a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + 5a_5 x^4 + \dots)$$

$$- \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots \right) (a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots) = 0$$

$$\Rightarrow (a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + 5a_5 x^4 + \dots)$$

$$- (a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots)$$

$$- (a_0 x + a_1 x^2 + a_2 x^3 + a_3 x^4 + \dots)$$

$$- \left(\frac{a_0}{2} x^2 + \frac{a_1}{2} x^3 + \frac{a_2}{2} x^4 + \dots \right)$$

$$- \left(\frac{a_0}{6} x^3 + \frac{a_1}{6} x^4 + \dots \right)$$

$$- \left(\frac{a_0}{24} x^4 + \dots \right) + \dots$$

$$\Rightarrow (a_1 - a_0) + (2a_2 - a_1 - a_0)x + (3a_3 - a_2 - a_1 - \frac{a_0}{2})x^2 \\ + (4a_4 - a_3 - a_2 - \frac{a_1}{2} - \frac{a_0}{6})x^3 \\ + (5a_5 - a_4 - a_3 - \frac{a_2}{2} - \frac{a_1}{6} - \frac{a_0}{24})x^4 + \dots = 0$$

$$\Rightarrow a_1 - a_0 = 0 \Rightarrow a_1 = a_0 \quad (\underline{a_0 = 1}) \\ \Rightarrow \underline{a_1 = 1}$$

$$\Rightarrow 2a_2 - a_1 - a_0 = 0 \Rightarrow a_2 = \frac{1}{2}a_1 + \frac{1}{2}a_0 = 1 \Rightarrow \underline{a_2 = 1}$$

$$\Rightarrow 3a_3 - a_2 - a_1 - \frac{a_0}{2} = 0 \Rightarrow 3a_3 = 1 + 1 + \frac{1}{2} \Rightarrow \underline{a_3 = \frac{5}{6}}$$

$$\Rightarrow 4a_4 = a_3 + a_2 + \frac{a_1}{2} + \frac{a_0}{6} = \frac{5}{6} + 1 + \frac{1}{2} + \frac{1}{6} \Rightarrow \underline{\frac{5}{8} = a_4}$$

$$\text{So } y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots \\ = 1 + x + x^2 + \frac{5}{6}x^3 + \frac{5}{8}x^4 + \dots$$

Alternatively: $y(0) = 1$, $y' = e^x y \Rightarrow y'(0) = e^0 y(0) = 1$

$$y'' = e^x y + e^x y' \Rightarrow y''(0) = e^0 y(0) + e^0 y'(0) = 2$$

$$y''' = e^x y + 2e^x y' + e^x y'' \Rightarrow y'''(0) = 5$$

$$y^{(4)} = e^x y + 3e^x y' + 3e^x y'' + e^x y''' \Rightarrow y^{(4)}(0) = 15$$

$$y = \sum_{n=0}^{\infty} \frac{y^{(n)}(0)}{n!} x^n = 1 + 1 \cdot x + \frac{2}{2!} x^2 + \frac{5}{6} x^3 + \frac{15}{24} x^4 + \dots$$